Evaluating the Impact of Bolsa Família on Human Development and Child Health: Evidence from Brazilian Municipalities

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1 Introduction

Brazil's Bolsa Família Program (BFP) has become one of the most extensively studied conditional cash transfer (CCT) schemes in the world. Since its creation in 2003 under the Lula administration — through the unification of four pre-existing programs (Bolsa Escola, Bolsa Alimentação, Cartão Alimentação, and Auxílio Gás) — it rapidly grew into the largest CCT program globally, reaching over 30 million beneficiaries (Hall, 2006).

The Brazilian economic context when BFP emerged (1st Lula's presidential mandate) was one that Morais and Saad-Filho (2011) characterizes as a "hybrid" model that combined a neo-developmentalist expansion of social spending with the orthodox macroeconomic framework inherited from the Cardoso era — the so-called *tripé* of inflation targeting, floating exchange rates, and primary budget surpluses. Paradoxically, this combination produced significant poverty reduction and GDP growth, at least until 2010, though the authors warn that the coexistence of fundamentally incompatible policy logics was inherently temporary. Barbosa-Filho (2008) offers a complementary reading of the first Lula term (2003–2006), showing how high real interest rates — which benefited wealthier segments — were paired with large upward adjustments to the minimum wage and rapid expansion of social transfers, resulting in a decline in the Gini coefficient from 0.573 to 0.548 and an estimated 7.4% of the population exiting poverty.

Hall (2006) provides a critical but balanced early assessment of the BFP. He acknowledges the program's administrative achievements — particularly the simplification brought about by unifying fragmented predecessor schemes — while raising substantive concerns about targeting errors, benefit duplication, clientelism, and weak enforcement of conditionalities at the municipal level. Most importantly, Hall argues that the BFP risks generating a culture of welfare dependency without addressing the structural roots of poverty: land

concentration, lack of decent employment, and persistent inequalities in education and health. The political instrumentalization of the program ahead of the 2006 elections is also documented. These early criticisms established a recurring tension in the literature between the program's measurable short-run impacts and its long-run structural adequacy.

Subsequent quantitative research has largely confirmed positive short- and medium-term effects. Soares et al. (2010) compare the BFP to analogous CCT programs in Mexico (Oportunidades), Colombia, Ecuador, and Chile, finding that it successfully reduced extreme poverty and income inequality, improved school attendance, and lowered dropout rates. Crucially, they find no negative effects on labor market participation — and even a positive labor supply response, particularly among women. Where the BFP underperformed relative to its stated objectives — particularly in health and nutrition — the evidence points to supply-side constraints in health infrastructure rather than lack of demand from beneficiaries. Ciula (2022) further corroborates these findings using a multidimensional poverty index (MPI) based on the Alkire-Foster methodology combined with difference-in-differences estimation. The analysis shows that the BFP reduced multidimensional poverty for both moderately poor and vulnerable households, with the largest gains in living standards and community life dimensions, though effects were more limited for the most deprived families.

On household expenditure, Ferrario (2014) uses propensity score matching on the 2008–2009 household budget survey and finds that BFP transfers were directed primarily toward food — including poultry, vegetables, cereals, and legumes — and, notably, school supplies. The absence of significant increases in expenditure on tobacco or alcohol challenges stigmatizing narratives about the spending behavior of transfer recipients. Female-headed households in particular exhibited a higher propensity to invest in children's education.

The relationship between conditional cash transfers and health remains an active area of inquiry. Pescarini et al. (2020) present a protocol for estimating the causal impact of the BFP on premature mortality (ages 30–69) from cardiovascular and all causes, exploiting the income eligibility threshold through a regression discontinuity design using data from the 100 Million Brazilians Cohort. Similarly, Falcão et al. (2022) propose a study assessing BFP effects on maternal and child health outcomes — including birth weight, preterm births, maternal mortality, and child growth — using propensity score matching on the same cohort. These protocol papers reflect a methodological shift toward quasi-experimental designs capable of providing more credible causal estimates.

In a broader regional perspective, Stampini et al. (2025) analyze non-contributory transfer systems across 17 Latin American and Caribbean countries and find persistent structural gaps: transfers cover only 33% of the poverty gap on average, and just 55% of the poor are reached by existing programs. Despite these system-wide limitations, Brazil (along with Suriname) is identified as having one of the best-designed social protection architectures in the region, suggesting that the institutional legacy of the BFP has contributed to a relatively robust national framework.

The literature converges on a picture of a program that has delivered meaningful improvements in income poverty, inequality, school attendance, and household consumption, while operating within a macroeconomic context that constrained deeper structural transformation. Key gaps remain in the causal identification of health impacts, particularly over longer time horizons.

The present study contributes to this literature by providing municipality-level panel evidence on two distinct dimensions of the program's impact: aggregate human development, as measured by the Firjan Municipal Development Index (IFDM), and child health, as measured by the infant mortality rate. The two outcomes are analysed through complementary two-way fixed effects and event study specifications, allowing us to assess both the average treatment intensity effect and its dynamic evolution over a twenty-year horizon.

2 Data and Variables

The present analysis is based on two separate estimation samples. The first is a balanced municipality-year panel covering 5,561 Brazilian municipalities over the period 2006–2016, for a total of 60,362 observations. The second covers 5,570 Brazilian municipalities (before sample restriction) over the period 2000–2020, for a total of 116,970 observations. We have two different models with slightly different specifications.

2.1 Outcome variables

The first outcome variable is the *Índice Firjan de Desenvolvimento Municipal* (IFDM), a composite index ranging from 0 to 1 published annually by the Firjan System that aggregates municipal-level performance in three dimensions: employment and income, education, and health. Higher values indicate better overall development outcomes. The index is treated as a continuous dependent variable and enters the regression in levels, as it is bounded and not amenable to logarithmic transformation.

The second outcome variable is the *infant mortality rate* (IMR), defined as the number of deaths occurring in the first year of life per 1,000 live births in municipality i in year t:

$$\text{IMR}_{it} = \frac{\text{InfantDeaths}_{it}}{\text{LiveBirths}_{it}} \times 1{,}000$$

Data on infant deaths are sourced from the Sistema de Informações sobre Mortalidade (SIM) and data on live births from the Sistema de Informações sobre Nascidos Vivos (SINASC), both administered through DATASUS. Following standard practice in the Brazilian health economics literature (Aquino et al., 2009; Rasella et al., 2013; Victora et al., 2011), municipalities with fewer than 20 live births in a given year are excluded from the estimation sample to prevent mechanically extreme mortality rates driven by small-denominator noise (e.g., a single death among five births would produce an IMR

of 200). When infant deaths are recorded as missing for a municipality with observed live births, deaths are imputed to zero, consistent with DATASUS reporting conventions whereby municipalities with no recorded deaths in a given year may not submit a return. [1] In addition, observations with IMR exceeding 300 per 1,000 live births are dropped as data-entry artefacts; biologically implausible rates of this magnitude arise when a single death and a single live birth are registered in different administrative years within the same municipality, a known recording defect in DATASUS (Victora et al., 2011).
Additionally, winsorization at the 99th percentile of the distribution is performed to deal with measurement errors in small municipalities, where the mismatch between the household-level beneficiary register and individual-level population denominators often results in coverage ratios exceeding one. [2]

Starting from an estimation sample of 5,553 municipalities over the period 2000–2020, yielding 116,613 municipality-year observations, after these reductions and the deletion of municipality-year cells with missing values in `ln(GDP per capita)` or `ln(population)` we end up with the 113,857 observations used in the Poisson model. This broader temporal window is advantageous for identification: the four calendar years preceding the national launch of *Bolsa Família* in 2004 (2000–2003) provide genuine pre-treatment variation that can be used to test the parallel trends assumption directly, without recourse to the de-trending correction required for the IFDM model.

2.2 Treatment variable

The treatment variable is *coverage*, defined as the share of the municipal population enrolled in the *Bolsa Família* program in a given year ($\text{coverage}_{it} = B_{it}/P_{it}$, where B_{it} denotes the number of beneficiary households and P_{it} the resident population), and is common to both models.[3] Data on beneficiaries at the municipal level are drawn from IPEADATA for both estimation samples.

The IFDM model utilizes a panel from 2006 to 2016, a period during which *Bolsa Família* (BF) coverage varied continuously across all municipalities. In contrast, the IMR model extends over a twenty-year horizon (2000–2020). For the latter, coverage is set to zero for the 2000–2003 period, representing structural zeros prior to the program's national launch. The event study specifications anchor treatment intensity to different reference years to

[1] In the raw data, 25,387 municipality-year observations have missing infant deaths, of which the overwhelming majority also have small population counts. The distribution of missingness across years shows no structural break coinciding with BF's launch, confirming that the pattern reflects administrative suppression of zero-death cells rather than true data loss.

[2] Values of coverage in excess of one are present in 218 municipality-year cells across 127 municipalities. Winsorization at p99 is preferred over a hard cap at one because the variable is conceptually an intensity proxy rather than a strict participation rate.

[3] Strictly, B_{it} counts beneficiary *households* while P_{it} counts *individuals*, so coverage constitutes an intensity proxy rather than a strict participation rate.

reflect these distinct windows: baseline coverage is fixed at 2006 for the IFDM model to avoid contamination from endogenous post-2006 variation [4], while the IMR model utilizes 2003 – the final year prior to program entry —- as the natural reference point to fully exploit the pre-treatment data.

2.3 Construction of Real GDP per Capita and Deflation

The construction of the economic control variable, $\ln(\text{GDP per capita})_{it}$, requires specific adjustments to ensure longitudinal consistency. Given the volatile inflationary environment in Brazil, where consumer prices (IPCA) increased approximately threefold between 2000 and 2020, we deflate all nominal values to constant 2010 prices (for IMDF model) and 2020 prices (for IMR model). This process is handled via the `deflateBR` package, which applies the IPCA index at the annual level. Without this deflation, the model would conflate nominal price increases with real welfare gains, severely biasing the coefficient estimates for income and the year fixed effects.

Beyond price adjustments, for the IMR model the extended time horizon necessitates the integration of two distinct national accounting series from the IBGE/SIDRA database:

- **2000–2001:** Data are sourced from the legacy national accounts (Table 21).
- **2002–2020:** Data are sourced from the updated series (Table 5938).

To ensure that the transition between these series does not introduce structural breaks, we verified the continuity by inspecting the distribution of log-GDP growth rates at the 2001–2002 junction. The growth rates at this point were found to be consistent with adjacent year-pairs, confirming that the series switch does not introduce measurement artifacts.

In both models, to calculate the per capita values, the deflated municipal GDP is divided by the resident population estimates. For intercensal years where official counts were missing (notably 2007 and 2012), we applied linear interpolation between the nearest available data points to ensure a smooth denominator and avoid artificial volatility in the income series. We can see in Table 1 the comparison of the variables.

[4] This is a maintained limitation of the IFDM event study specification: since Bolsa Família was already operational and near-universally adopted across municipalities by 2006, the baseline year does not represent a genuine pre-treatment state. The choice of 2006 as the reference year is driven entirely by data availability, as the IFDM series does not extend prior to that year. Among the available years, 2006 corresponds to the lowest average coverage intensity before the large-scale program expansion, making it the most plausible baseline. The implications of this choice are discussed further in the Limitations section.

Table 1: Variables and Data Sources

Variable	Description	Period	Source
Panel A — IFDM model			
IFDM	Municipal development index (0–1).	2006–2016	Firjan
coverage_{it}	Share of population enrolled in *Bolsa Família*.	2006–2016	IPEADATA
$\ln(\text{GDP pc})_{it}$	Log of municipal nominal GDP per capita (deflated to 2010 BRL via IPCA index).	2006–2016	IBGE/SIDRA
$\ln(P)_{it}$	Log of municipal population.	2006–2016	IBGE/TCU
Panel B — IMR model			
IMR_{it}	Infant deaths per 1,000 live births.	2000–2020	DATASUS
coverage_{it}	Share of population enrolled in *Bolsa Família* (set to 0 for 2000–2003).	2000–2020	IPEADATA
$\ln(\text{GDP pc})_{it}^{\text{real}}$	Log of real GDP per capita (deflated to 2020 BRL via IPCA index).	2000–2020	IBGE/SIDRA
$\ln(P)_{it}$	Log of municipal population.	2000–2020	IBGE

3 Methodology

3.1 Two-Way Fixed Effects (TWFE)

To estimate the effect of the *Bolsa Família* program on municipal development outcomes and child health, we adopt a *Difference-in-Differences* (DiD) framework with a continuous treatment intensity, implemented through a *Two-Way Fixed Effects* (TWFE) estimator. Unlike the canonical binary DiD, the continuous-treatment TWFE exploits cross-sectional variation in program intensity — measured by the share of the population receiving benefits (*coverage*) — together with its variation over time, while simultaneously controlling for time-invariant unobserved heterogeneity at the municipal level and for aggregate shocks common to all municipalities.

The baseline specification (first model) is:

$$y_{it} = \beta\, \text{coverage}_{it} + \mathbf{X}_{it}'\boldsymbol{\gamma} + \alpha_i + \lambda_t + \varepsilon_{it} \tag{1}$$

where y_{it} is the outcome of municipality i in year t, $\text{coverage}_{it} = B_{it}/P_{it}$ is the share of the population enrolled in *Bolsa Família*, $\mathbf{X}_{it}$ is a vector of time-varying controls comprising the log of GDP per capita and the log of population, α_i is a municipality fixed effect that absorbs all time-invariant structural characteristics, λ_t is a year fixed effect common to all municipalities, and ε_{it} is an error term clustered at the municipal level to account for serial autocorrelation. The coefficient of interest is β: since y_{it} is in levels, it measures the absolute change in the outcome associated with a one-percentage-point increase in coverage.

For the infant mortality outcome, the baseline specification in equation (1) is replaced by a

Poisson pseudo-maximum likelihood (PPML) estimator (Silva and Tenreyro, 2006; Rasella et al., 2013):

$$\mathbb{E}\big[\,\mathrm{IMR}_{it} \mid X_{it}, \alpha_i, \lambda_t\big] = \exp\big(\beta\,\mathrm{coverage}_{it} + X_{it}'\gamma + \alpha_i + \lambda_t\big) \tag{2}$$

where all variables are defined as in equation (1).
The PPML estimator is preferred to log-linear OLS because the distribution of IMR departs substantially from normality: approximately 22,000 municipality-year cells record zero infant deaths, rendering log transformation either undefined or sensitive to the arbitrary choice of offset constant. PPML accommodates zeros naturally and does not require distributional assumptions beyond correct specification of the conditional mean. Log-linear OLS specifications are retained as robustness checks. Fixed effects and clustered standard errors follow the same structure as equation (1).

3.2 Identification and Parallel Trends

Causal identification of β requires that, in the absence of the program, municipalities with different treatment intensities would have followed parallel outcome trajectories (*parallel trends assumption*). For the IFDM model, the absence of pre-2006 data prevents a direct test of this assumption. We instead adopt an event study specification in which treatment intensity is fixed at its 2006 baseline value, $c_i \equiv \mathrm{coverage}_{i,2006}$, to prevent endogenous variation in coverage from contaminating the estimated coefficients:

$$\mathrm{IFDM}_{it} = \sum_{\tau \neq 2006} \beta_\tau \left(\mathbf{1}[t=\tau] \times \mathrm{coverage}_{i,2006}\right) + \mathbf{X}_{it}'\boldsymbol{\gamma} + \alpha_i + \lambda_t + \varepsilon_{it} \tag{3}$$

where $\mathrm{coverage}_{i,2006}$ is the baseline coverage rate of municipality i in 2006, held fixed across years. The reference year is 2006, so each β_τ captures the differential IFDM trajectory of high-coverage municipalities relative to low-coverage ones, compared to the 2006 baseline. Since the program was already operational in 2006, these coefficients do not recover effects from zero exposure; rather, they capture the *evolution* of the outcome differential relative to an already-treated baseline.

The raw event study coefficients increase monotonically from 2007 onward, which could reflect either a genuine causal effect of the program or the continuation of a pre-existing differential trend that predates our observation window. To partially address this ambiguity, we estimate a *de-trended* version of the event study. Specifically, we estimate the municipality-specific interaction between year and baseline coverage — allowing for a quadratic time profile — on the 2006–2011 sub-period, and subtract the fitted trend from the outcome before re-estimating equation (3).

This procedure carries an important limitation that must be stated explicitly. The de-trending window (2006–2011) is not a genuine pre-treatment period: *Bolsa Família* was already active and expanding throughout those years. To the extent that the program itself generated differential IFDM improvements during 2006–2011, subtracting the estimated trend removes variation that is partly causal, not merely confounding. As a result, the de-trended coefficients for the post-2011 period should be interpreted as capturing the *acceleration* of the treatment effect relative to the 2006–2011 trajectory, rather than the total effect from zero exposure. The identifying assumption is therefore that the post-2011 surge in the de-trended coefficients — coinciding with the large-scale programme expansion under *Brasil sem Miséria* — reflects a genuine intensification of the causal effect, over and above both pre-existing trends and any smooth continuation of the 2006–2011 dynamics.

Moving to the analysis of the second model, the identifying variation in equation (4) is cross-municipal: municipalities that happened to have higher BFP penetration at the moment of national rollout in 2004 serve as an implicit high-intensity treatment group, while those with lower initial coverage serve as a comparison group. The event study recovers the differential IMR trajectory of high-intensity municipalities relative to low-intensity ones, year by year, normalised to their common pre-program baseline. Under the parallel trends assumption — which the pre-treatment coefficients for 2000–2002 are used to test directly — a post-2004 divergence in $\hat{\delta}_s$, growing in absolute magnitude and statistically significant, constitutes evidence of a causal protective effect of program intensity on infant mortality. No pre-existing differential trend is detectable in the years preceding the program: had high-intensity municipalities been on a systematically faster downward IMR trajectory already before 2004, we would expect $\hat{\delta}_{2000}$, $\hat{\delta}_{2001}$, and $\hat{\delta}_{2002}$ to be negative and significant, which Table 5 shows is not the case.

To capture this, the event study in equation (3) is re-specified with a Poisson conditional mean. The design follows a dose-response logic in which treatment intensity is fixed at each municipality's initial scale of program operation, anchoring $c_i \equiv \text{coverage}_{i,2004}$ to the first year of national rollout rather than to 2006 as in the IFDM specification. The year 2003 – the last pre-treatment year – serves as the temporal reference point ($\delta_{2003} = 0$ by construction):

$$\mathbb{E}[\,\text{IMR}_{it} \mid \cdot\,] = \exp\left(\sum_{s \neq 2003} \delta_s \big(c_i \times \mathbf{1}[t = s]\big) + \mathbf{X}'_{it}\boldsymbol{\gamma} + \alpha_i + \lambda_t \right) \tag{4}$$

The coefficients δ_s are interpreted as the semi-elasticity of IMR with respect to baseline BF intensity in year s relative to 2003: $(e^{\delta_s} - 1) \times 100$ gives the approximate percentage difference in expected IMR between a municipality with one additional unit of baseline coverage and one with zero, in year s compared to the pre-treatment reference. The availability of four genuine pre-treatment years (2000–2003) means that δ_s for $s \in \{2000, 2001, 2002\}$ serve

as direct falsification tests without any de-trending correction: under the parallel trends assumption, these coefficients should be statistically indistinguishable from zero.

It bears emphasis, however, that the baseline coverage in 2004 is not randomly assigned. Municipalities that enrolled more families early were disproportionately poorer, had worse baseline health infrastructure, and may have had more active local social services administrations. The parallel trends assumption requires that none of these characteristics generated a differential *trend* in IMR relative to lower-intensity municipalities over the 2000–2003 period — a weaker condition than requiring identical levels, and one that the pre-trend evidence supports. Nevertheless, the design should be interpreted as providing *strong observational evidence* of a dose-response relationship between BFP intensity and infant mortality reduction, rather than a sharp causal estimate in the sense of an experimental or discontinuity-based design.

A maintained assumption underlying both specifications is that, conditional on municipality and year fixed effects, within-unit changes in BF coverage are uncorrelated with idiosyncratic shocks to the outcome variable. Municipality fixed effects absorb time-invariant confounders such as chronic institutional capacity or persistent poverty culture, while year fixed effects absorb aggregate macroeconomic shocks. The residual concern is time-varying selection: municipalities that simultaneously improve their social services enrollment capacity and experience independent improvements in health or development outcomes would generate upward bias in the estimated $\hat{\beta}$. We address this partially through the event study design, which anchors treatment intensity to its baseline value and therefore is not contaminated by post-treatment changes in coverage driven by outcome feedback. The inclusion of log real GDP per capita as a time-varying control further absorbs the local economic cycle component of this potential confounding.

4 Results

4.1 Baseline TWFE Estimates

For the first model, Table 2 reports the results of the three TWFE specifications. In the baseline model without controls, the estimated coefficient on *coverage* is positive but statistically insignificant ($\hat{\beta} = 0.00071$, $p = 0.125$), suggesting that a naive specification conflates the effect of the program with correlated time-varying confounders. Once log GDP per capita and log population are included, the point estimate on coverage remains positive and similarly sized ($\hat{\beta} = 0.00044$) but still falls short of conventional significance thresholds ($p = 0.303$). By contrast, log GDP per capita enters with a precisely estimated coefficient of 0.027 ($p < 0.001$), indicating that a 1% increase in local per capita income is associated with an increase of approximately 0.027 points in the IFDM, a substantively meaningful effect given the index's 0–1 range. Log population is not statistically distinguishable from zero in either specification.

Table 2: Main Results: TWFE Effect of Bolsa Família on IFDM (2006–2016)

	Baseline	With Controls	Semi-Log
BF coverage	0.001	0.000	
	(0.000)	(0.000)	
ln(BF coverage)			0.002***
			(0.000)
ln(Real GDP p.c.)		0.027***	0.027***
		(0.002)	(0.002)
ln(Population)		−0.001	0.001
		(0.004)	(0.004)
Municipality FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Num.Obs.	60 362	60 362	60 362

* p <0.1, ** p <0.05, *** p <0.01

The semi-log specification, which replaces the level of coverage with its natural logarithm, yields a markedly different picture: the coefficient on ln(coverage) is 0.0024 and highly significant ($p < 0.001$), implying that a 1% increase in the share of the population enrolled in *Bolsa Família* is associated with an increase of approximately 0.0024 points in the IFDM. This result suggests that the relationship between program intensity and development outcomes is better captured on a proportional scale, consistent with diminishing marginal returns to coverage at higher enrollment rates.

For the second model, Table 3 reports Poisson TWFE estimates with raw IMR as the dependent variable. In the baseline specification, the coefficient on coverage is $\hat{\beta} = -0.147$ ($p = 0.085$), negative and marginally significant at the 10% level. Adding log real GDP per capita and log population yields $\hat{\beta} = -0.132$ ($p = 0.136$), which remains negative and of similar magnitude, though precision falls slightly.

Comparing the Poisson model with the OLS specifications reveals a critical sensitivity to the functional form and the handling of zero-valued observations. In the OLS models ($\ln(y+1)$ and $\ln(y+0.1)$), the population coefficient is large and highly significant ($p < 0.01$), and the estimated impact of *Bolsa Família* flips sign. This pattern suggests a significant recording bias: in smaller or developing municipalities, population growth and administrative expansion are often proxies for the improved completeness of death registration within

Table 3: Main Results: Poisson TWFE Effect of Bolsa Família on IMR

	Baseline	With Controls	Log-Log
BF intensity	−0.147*	−0.132	
	(0.085)	(0.087)	
ln(BF intensity)			−0.008
			(0.005)
ln(Real GDP p.c.)		−0.014	−0.014
		(0.015)	(0.015)
ln(Population)		0.013	0.014
		(0.026)	(0.026)
Num.Obs.	113 857	113 857	113 857

* p <0.1, ** p <0.05, *** p <0.01

the DATASUS system. OLS misinterprets this increase in documented cases as a genuine rise in mortality, effectively confounding data quality with program impact. In contrast, the Poisson model accounts for the structural zeros without such distortions, yielding a non-significant population coefficient ($p = 0.800$) and ensuring that the results are not contaminated by administrative artifacts.

While these TWFE results are point-estimated in the direction predicted by theory, the lack of precision is consistent with the identification challenge: since *Bolsa Família* rollout was largely monotonic within municipalities, there is limited temporal variation to exploit once unit and year fixed effects are removed. This reinforces the necessity of the event study approach, which leverages cross-municipal variation in baseline program intensity to more effectively detect the treatment effect.

4.2 Event Study and Parallel Trends

For the first model, Table 4 displays the event study coefficients $\hat{\delta}_s$ from equation (3), where treatment intensity is anchored to baseline coverage in 2006. In the raw event study, the coefficients increase monotonically from 2007 onward — rising from 0.273 in 2007 to 0.795 in 2016, all significant at the 1% level — reflecting a smooth upward trend in the IFDM that is proportional to initial programme intensity. The absence of a pre-period to test against (2006 is the reference year) motivates the de-trending procedure.

After removing the pre-existing quadratic trend estimated on the 2006–2011 sub-period (with $\hat{\beta}_{\text{lin}} = 0.098$ and $\hat{\beta}_{\text{quad}} = -0.024$), the de-trended coefficients for 2009 and 2010

become statistically indistinguishable from zero ($\hat{\delta}_{2009} = 0.032$, $p = 0.115$; $\hat{\delta}_{2010} = 0.002$, $p = 0.914$), providing support for the parallel trends assumption in those years. The coefficient for 2007 remains significant ($\hat{\delta}_{2007} = 0.103$, $p < 0.001$), suggesting some residual pre-trend in the very first year, while 2011 shows a moderate positive value ($\hat{\delta}_{2011} = 0.079$, $p = 0.001$) that may reflect the onset of the program's large-scale expansion. From 2012 onward, the de-trended coefficients accelerate sharply, growing from 0.191 in 2012 to 1.241 in 2016, all significant at the 1% level. This post-2011 surge — coinciding with the major expansion of *Bolsa Família* under the *Brasil sem Miséria* plan — is interpreted as evidence of a causal effect of the program on municipal development outcomes over and above any pre-existing trend.

The de-trending procedure is motivated by the need to separate two potentially overlapping dynamics. In the raw event study, coefficients rise monotonically from 2007 to 2016, which could reflect either a genuine causal effect of the programme or a pre-existing differential trend — municipalities with higher baseline coverage may have been improving faster on the IFDM for reasons unrelated to *Bolsa Família*, such as structural characteristics correlated with early programme adoption. Without a genuine pre-treatment period, the two explanations are observationally equivalent. The de-trending procedure addresses this by estimating the differential trend over 2006–2011 and removing it from the outcome. What remains is the component of the IFDM trajectory that cannot be explained by that pre-existing dynamic. The fact that the de-trended coefficients for 2009 and 2010 are statistically indistinguishable from zero confirms that the estimated trend fits those years well, while the sharp acceleration from 2012 onward — growing from $\hat{\delta}_{2012} = 0.191$ to $\hat{\delta}_{2016} = 1.241$ — is incompatible with a smooth continuation of the 2006–2011 trajectory and coincides precisely with the large-scale programme expansion under *Brasil sem Miséria*.

For the second model, the Poisson event study coefficients provide markedly cleaner identification than the log-linear alternatives. As we can see in figure 1, all three pre-treatment coefficients (2000–2002) are statistically indistinguishable from zero, with the 2000 point estimate slightly positive (from table 5, $\hat{\delta}_{2000} = 0.049$, $p = 0.902$). This constitutes direct evidence in favor of the parallel trends assumption: municipalities that would eventually receive higher BF intensity show no differential pre-program trajectory in infant mortality. Post-2004, coefficients turn consistently negative and grow in absolute magnitude, reaching statistical significance from 2007 onward ($\hat{\delta}_{2007} = -1.43$, $p = 0.001$) and peaking at $\hat{\delta}_{2014} = -2.43$ ($p < 0.001$). The effect moderates in 2020 ($\hat{\delta}_{2020} = -1.22$, $p = 0.013$), consistent with COVID-19 disruption to both healthcare delivery and administrative reporting. For magnitude: a one standard-deviation increase in baseline coverage ($\sigma = 0.034$) is associated with a $1 - e^{-0.034 \times 2.43} \approx 7.9\%$ reduction in expected IMR by 2014, or roughly 1.3 fewer infant deaths per 1,000 live births at the sample mean. These findings reinforce the conclusion that *Bolsa Família* serves as a structural pillar for public health in Brazil, with its effectiveness scaling directly with the intensity and duration of municipal coverage.

Table 4: Event Study: Effect of Bolsa Família on IFDM by Year

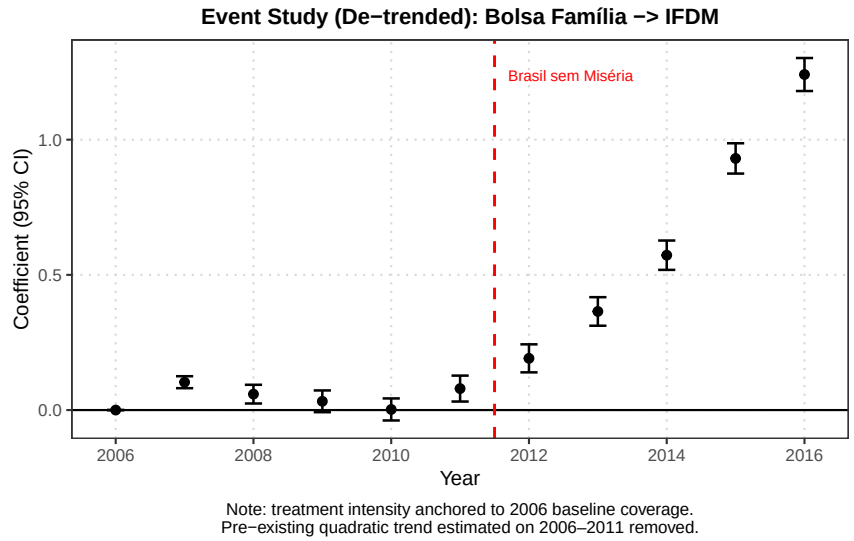
Year	Estimate	Low (95%)	High (95%)	p-value
Reference Year				
2006	0.000	–	–	–
Original Event Study				
2007	0.273	0.251	0.295	0.000
2008	0.351	0.316	0.385	0.000
2009	0.398	0.358	0.438	0.000
2010	0.395	0.354	0.436	0.000
2011	0.452	0.404	0.500	0.000
2012	0.495	0.443	0.547	0.000
2013	0.552	0.499	0.605	0.000
2014	0.597	0.543	0.651	0.000
2015	0.743	0.687	0.799	0.000
2016	0.795	0.734	0.856	0.000
De-trended Event Study				
2007	0.103	0.081	0.125	0.000
2008	0.059	0.024	0.093	0.001
2009	0.032	-0.008	0.073	0.115
2010	0.002	-0.038	0.043	0.914
2011	0.079	0.032	0.127	0.001
2012	0.191	0.139	0.243	0.000
2013	0.365	0.312	0.418	0.000
2014	0.573	0.518	0.627	0.000
2015	0.931	0.875	0.987	0.000
2016	1.241	1.180	1.301	0.000

Note: Estimates represent the differential IFDM of municipalities with a one-unit higher baseline (2006) coverage rate, relative to the reference year 2006. The de-trended specification removes a municipality-specific quadratic trend in coverage intensity estimated over 2006–2011 before re-estimating the event study. 95% confidence intervals and p-values are based on standard errors clustered at the municipal level. Observations: 58,226. Municipality and year fixed effects included in all specifications.

Table 5: Poisson Event Study: Coefficients by Year

Year	Estimate	Low (95%)	High (95%)	p-value
Pre-treatment				
2000	0.049	-0.734	0.833	0.902
2001	-0.654	-1.514	0.206	0.136
2002	-0.642	-1.431	0.147	0.111
Reference Year				
2003	0.000	–	–	–
Post-treatment				
2004	-0.387	-1.130	0.357	0.308
2005	-0.912	-1.736	-0.089	0.030
2006	-0.705	-1.506	0.097	0.085
2007	-1.428	-2.256	-0.599	0.001
2008	-1.855	-2.729	-0.981	0.000
2009	-1.631	-2.480	-0.781	0.000
2010	-1.717	-2.582	-0.852	0.000
2011	-1.843	-2.739	-0.948	0.000
2012	-1.799	-2.699	-0.899	0.000
2013	-1.184	-2.090	-0.278	0.010
2014	-2.434	-3.324	-1.543	0.000
2015	-2.135	-3.105	-1.165	0.000
2016	-2.143	-3.052	-1.235	0.000
2017	-1.490	-2.386	-0.594	0.001
2018	-2.139	-3.085	-1.193	0.000
2019	-2.395	-3.361	-1.428	0.000
2020	-1.220	-2.184	-0.256	0.013

Note: Estimates represent semi-elasticities from a Fixed-Effects Poisson model, which accounts for the high frequency of zero-valued observations in the infant mortality data. Standard errors are clustered at the municipality level.



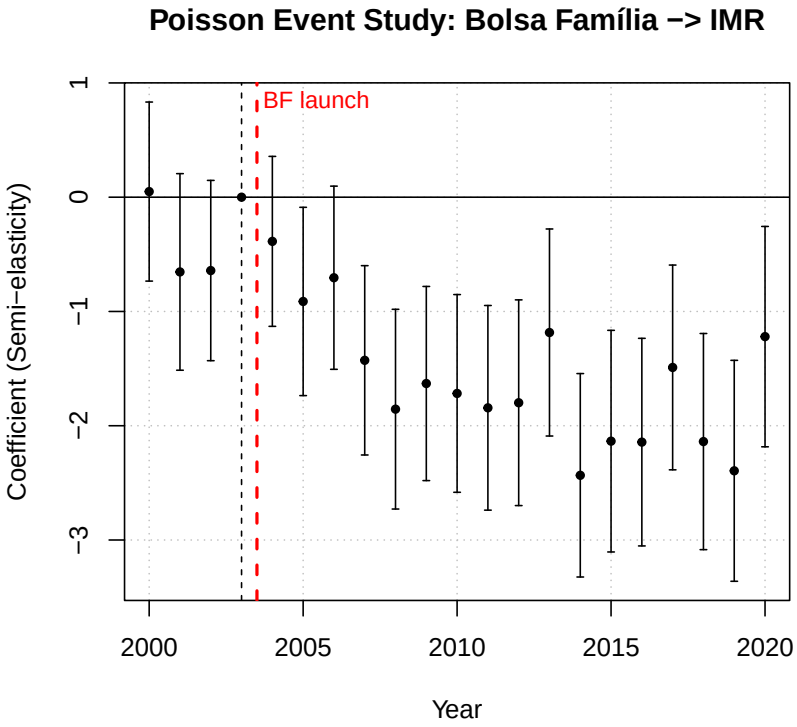


Figure 1: Event study results: IMR (top) and IFDM de-trended (bottom)

5 Limitations

Looking at the study as a whole, the two models have some common limits, in particular with regards to endogeneity of the coverage variable. Municipal enrolment in *Bolsa Família* is determined by the interaction of household poverty, local administrative capacity, and targeting effort — all of which may independently affect the outcomes under study. Municipality fixed effects α_i absorb the component of this selection that is time-invariant: structural poverty levels, geographic remoteness, and chronic institutional capacity differences are differenced out. However, time-varying confounders remain a threat. Municipalities that simultaneously improve their social services, expand their civil registry infrastructure, or invest in primary healthcare will both enrol more *Bolsa Família* beneficiaries and achieve better health and development outcomes independently of the cash transfer itself.

In the absence of a credible instrument for coverage, our identification rests on the following maintained assumption: *conditional on municipality and year fixed effects, changes in coverage are uncorrelated with idiosyncratic shocks to municipal outcomes.* This assumption would be violated if, for example, federal targeting reforms differentially expanded enrolment in municipalities that were already on a steeper improvement trajectory. We cannot formally test this condition, but the event-study patterns — in particular the absence of pre-trend acceleration after de-trending — are consistent with it.

5.1 IFDM model

A major limitation of this model concerns the absence of a genuine pre-treatment period. The *Bolsa Família* programme was launched in 2003, whereas our municipal panel begins in 2006, the earliest year for which IFDM data are available. As a result, the baseline year coincides with a period in which the programme was already operational and coverage was already strictly positive in the vast majority of municipalities. The event-study specification uses 2006 as the reference year, so all estimated β_τ are interpreted relative to an already-treated baseline rather than a genuine counterfactual of zero exposure.

The quadratic de-trending procedure described in Section 3.2 partially compensates for this by removing pre-existing heterogeneous trends from the outcome variable. However, the trend parameters are estimated over 2006–2011, a sub-period during which *Bolsa Família* was already active and expanding. To the extent that programme exposure itself drove differential trends in IFDM during that window, the de-trending procedure removes variation that is partly causal, potentially attenuating the post-2011 estimates. We flag this as a maintained limitation: the identifying variation in the event study captures the *acceleration* of the treatment effect relative to the 2006 trajectory, not the effect from zero exposure.

5.2 IMR model

Few additional limitations of the infant mortality analysis require acknowledgement. First, the two-way fixed effects estimator identifies the effect of *Bolsa Família* from within-municipality variation in coverage over time. Since programme rollout was largely monotonic after initial entry – most municipalities expanded continuously once enrolled – the within-unit temporal variation available to the TWFE is limited once municipality and year fixed effects are removed. This explains the imprecision of the baseline Poisson TWFE estimates and is the reason the event study, which instead exploits cross-sectional variation in baseline programme intensity, constitutes the primary identification exercise.

Second, the infant mortality measure is subject to well-documented under-reporting in DATASUS, particularly in smaller and more remote municipalities in the earlier years of the panel. While our imputation of zero deaths for municipalities with observed live births but missing death records is consistent with standard practice in the Brazilian health economics literature, it introduces measurement error that is likely correlated with municipality size and institutional capacity — both of which are themselves correlated with programme intensity.

These limitations suggest that the results presented here, while consistent with a protective causal effect of *Bolsa Família* on human development and child health, should be interpreted as indicative rather than definitive. Credibly identifying the full range of the program's health and development impacts — across mortality, nutrition, education, and household welfare simultaneously — would require richer quasi-experimental designs.

6 Conclusions

The results for the IFDM model paint a consistent picture across specifications, albeit with important nuances. The baseline TWFE estimates in levels yield a positive but statistically insignificant coefficient on *Bolsa Família* coverage, both with and without controls ($\hat{\beta} = 0.0007$, $p = 0.125$; $\hat{\beta} = 0.0004$, $p = 0.303$). This imprecision is consistent with a non-linear relationship between programme intensity and development outcomes: when coverage enters in logarithmic form, the coefficient becomes highly significant ($\hat{\beta} = 0.0024$, $p < 0.001$), suggesting diminishing marginal returns to coverage at higher enrolment rates. A 1% increase in the share of the population enrolled in *Bolsa Família* is associated with an increase of approximately 0.0024 points in the IFDM — a meaningful effect given the index's 0–1 range and the sustained nature of the program.

The event study constitutes the primary identification exercise. Anchoring treatment intensity to its 2006 baseline value and removing the pre-existing quadratic differential trend estimated over 2006–2011, the de-trended coefficients for 2009 and 2010 are statistically indistinguishable from zero ($\hat{\delta}_{2009} = 0.032$, $p = 0.115$; $\hat{\delta}_{2010} = 0.002$, $p = 0.914$), consistent

with the parallel trends assumption over those years. From 2012 onward, coinciding with the large-scale expansion of *Bolsa Família* under the *Brasil sem Miséria* plan (launched in 2011 as an extension of the programme targeting extreme poverty), the de-trended coefficients accelerate sharply and monotonically, growing from $\hat{\delta}_{2012} = 0.191$ to $\hat{\delta}_{2016} = 1.241$, all significant at the 1% level. This post-2012 surge is interpreted as evidence of a causal effect of *Bolsa Família* on municipal development outcomes over and above any pre-existing differential trend, conditional on the identifying assumptions discussed in Section 5.

The magnitude of the effect is economically meaningful. By 2016, a municipality at the 75th percentile of baseline coverage experiences an IFDM trajectory approximately ($\hat{\delta}_{2016} \times \Delta c$) points higher than one at the 25th percentile, where Δc is the interquartile range of baseline coverage. Log GDP per capita enters with a precisely estimated coefficient of 0.027 ($p < 0.001$) across all specifications, confirming that local economic conditions are an independent determinant of municipal development outcomes. Log population is not statistically distinguishable from zero in any specification.

As for what concerns the second model, examining the effect of Bolsa Família intensity on infant mortality, the baseline Poisson specification yields a negative and marginally significant coefficient on coverage ($\hat{\beta} = -0.147$, $p = 0.085$), which remains negative and of similar magnitude once log real GDP per capita and log population are added ($\hat{\beta} = -0.132$, $p = 0.136$), though precision declines slightly. The attenuation of the coefficient after conditioning on controls is consistent with limited within-municipality temporal variation in BF intensity following the largely monotonic post-2004 rollout, a structural identification challenge common to continuous-treatment DiD designs applied to programs with near-universal take-up.

The event study resolves this identification challenge by exploiting cross-sectional variation in baseline program intensity. The three pre-treatment coefficients (2000–2002) are statistically indistinguishable from zero, providing direct support for the parallel trends assumption: municipalities destined to receive higher BF coverage show no differential pre-program trajectory in infant mortality. Post-2004 coefficients turn consistently negative and grow in absolute magnitude, achieving statistical significance from 2007 onward and peaking at $\hat{\delta}_{2014} = -2.43$. A one standard-deviation increase in baseline coverage ($\sigma \approx 0.034$) is associated with approximately a 7.9% reduction in expected IMR by 2014, or roughly 1.3 fewer infant deaths per 1,000 live births at the sample mean. The modest attenuation in 2020 is consistent with the disruption of primary healthcare services and administrative reporting by the COVID-19 pandemic rather than a reversal of the program's long-run protective effect.

Taken together, the two models provide complementary evidence on the reach of *Bolsa Família* across distinct dimensions of human welfare. The IFDM model documents a positive and increasing effect of program intensity on aggregate municipal development — a composite measure encompassing employment, education, and health — that emerges clearly

from 2012 onward once pre-existing differential trends are accounted for. The IMR model establishes that this aggregate improvement is accompanied by a substantive reduction in infant mortality, with effects growing in magnitude over the post-2004 period and reaching a peak reduction of approximately 7.9% in expected IMR per standard-deviation increase in baseline coverage by 2014.

The two results are mutually reinforcing. The IFDM captures a broad developmental trajectory that includes, but is not limited to, health outcomes; the IMR analysis isolates one of the most sensitive indicators of child welfare and confirms that the program's protective effect operates through channels — improved nutrition, uptake of preventive care induced by conditionality requirements, and reduced poverty-related stress on maternal health — that are conceptually distinct from the income and education dimensions driving the IFDM. That both outcomes respond positively to *Bolsa Família* intensity, across different estimation windows, different estimators, and different identification strategies, strengthens the overall credibility of the findings.

Several caveats apply jointly to both models. The identifying variation is cross-sectional in the event study specifications: municipalities with higher baseline programme intensity are compared to those with lower intensity, conditional on common time shocks and municipality fixed effects. While the pre-treatment evidence supports the parallel trends assumption — directly for the IMR model through the 2000–2003 pre-programme window, and indirectly for the IFDM model through the de-trending procedure — the possibility that baseline coverage correlates with unobserved time-varying determinants of both outcomes cannot be entirely ruled out. The inclusion of log real GDP per capita as a time-varying control partially mitigates this concern but does not eliminate it. These results should therefore be interpreted as robust within-municipality estimates of the association between *Bolsa Família* intensity and human development outcomes, consistent with a causal interpretation but not immune to residual confounding.

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